

# PATRICK KOLLER

PhD Candidate

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## OVERVIEW

PhD candidate working on scientific machine learning, focusing on physics-informed neural networks and neural operators for accelerating large-scale physical simulations. Backed by over 10 years of hands-on industry experience in algorithm and software development, machine learning, and project management, I am driven to bridge the gap between cutting-edge research and real-world applications.

## EDUCATION

- 09/2024 - Today **PhD Candidate in Electrical and Computer Engineering** **Northwestern University**
- PhD student in Prof. Dr. Katsaggelos's Image and Video Processing Lab (IVPL). GPA: 3.9/4.0
  - Conducting research in scientific machine learning, focusing on physics-informed neural networks (PINNs) and neural operators (PINOs) for accelerating large-scale physical simulations in astrophysics.
  - Developing machine learning methods for solving differential equations and modeling complex physical systems, with an emphasis on generalization, scalability, and physical consistency.
  - Research supported by the NSF-Simons AI Institute for the Sky (SkAI), leveraging large-scale HPC systems (e.g., *Delta*, *DeltaAI*, and *Northwestern's Quest HPC cluster*) for simulation and model development.
  - Previous work on explainable AI and robustness in deep learning, including methods for analyzing bias and improving model interpretability.
- 06/2026 - 07/2026 **Management for PhDs** **Kellogg School of Management - Northwestern University**
- Selected for an intensive six-week summer certificate program for post-candidacy doctoral students, taught by Kellogg faculty and based on content from the Kellogg MBA curriculum.
  - Studied management, leadership, strategy, finance, accounting, marketing, risk and uncertainty, and commercialization of research.
- 02/2020 - 02/2023 **Master of Science in Data Science** **Eastern Switzerland University of Applied Sciences**
- Part-time studies alongside working as an engineer at Stettbacher Signal Processing AG. Grade: 5.52/6
  - Master Thesis: Development of a novel explainable AI method called "Caption-based explainable AI" at the Image and Video Processing Lab at Northwestern University under the supervision of Prof. Dr. Katsaggelos, Prof. Dr. Guido Schuster and my mentor Amil Dravid. Grade: 6/6
  - Development of a quantitative trading engine using a combination of fixed horizon and triple barrier labels, gaussian processes to accelerate the hyperparameter tuning and a novel trading strategy. Grade: 5.5/6
  - Development of an "AI in injection molding" anomaly detection model, which is able to classify the quality of produced goods using the cavity pressure curve. We optimized a wide range of machine learning models like linear regression, autoencoder, variational-autoencoder, convolutional-autoencoder, gated-recurrent-unit-autoencoder, elliptic envelope, isolation forest, local outlier factor, and one-class support vector machines on a NVIDIA DGX-2. An autoencoder based model delivered the best performance considering the restricted resources of the embedded system. The success of this thesis led to the implementation in a product called ComoNeo by Kistler AG. Grade: 5.5/6
- 05/2014 **Vocational Trainer for Automation Engineers** **Berufsbildner AG**
- 09/2012 - 09/2016 **Bachelor of Science in Electrical Engineering** **Zurich University of Applied Sciences**
- Part-time studies alongside working as an automation engineer at Comsys Bärtsch AG.
  - Bachelor thesis: Development of electronics, optics, mechanics, and software for a medical device to detect brucellosis. A lock-in algorithm allows to detect a signal hidden in the noise. Grade: 5.5/6
  - Development of a force control system in a stranding machine to prevent a corkscrew effect during custom cable production. The project has been rewarded with a 1'500.00 CHF prize by the company Huber+Suhner AG. Grade: 5.5/6
- 08/2007 - 08/2011 **Apprenticeship as an Automation Engineer** **Comsys Bärtsch AG**

## PUBLICATIONS

- 04/2026 **Physics-Informed Neural Operators for Stellar Structure Modeling in Equilibrium** **Machine Learning: Science and Technology (under review)**  
S. Lopez-Tapia, M. Ballester, S. Gossage, P. M. Srivastava, U. Demir, **P. Koller**, C. Wuersch, S. Chakraborty, V. Kalogera, A. K. Katsaggelos

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| 04/2026 | <b>Learning the Stellar Structure Equations via Self-Supervised Physics-Informed Neural Networks</b><br>M. Ballester, S. Lopez-Tapia, S. Gossage, <b>P. Koller</b> , P. M. Srivastava, U. Demir, Y. Jo, A. P. Marquez, C. Wuer-sch, S. Chakraborty, V. Kalogera, A. K. Katsaggelos           | Scientific Reports (under review) |
| 03/2026 | <b>Irregularly Sampled Time Series Interpolation for Binary Evolution Simulations Using Dynamic Time Warping</b><br>U. Demir, P. M. Srivastava, A. K. Katsaggelos, V. Kalogera, S. Lopez-Tapia, M. Ballester, S. Lalvani, <b>P. Koller</b> , J. J. Andrews, S. Gossage, M. M. Briel, E. Teng | The Astrophysical Journal         |
| 09/2025 | <b>Caption-Driven Explainability: Probing CNNs for Bias via CLIP (Workshop)</b><br><b>P. Koller</b> , A. Dravid, G. M. Schuster, A. K. Katsaggelos                                                                                                                                           | IEEE ICIP                         |
| 09/2025 | <b>Size-Informed Representations for Unsupervised Image Clustering (Workshop)</b><br>P. R. M. Srivastava, C. Apostolidis, S. Pailwan, <b>P. Koller</b> , L. Stefanopoulos                                                                                                                    | IEEE ICIP                         |

## AWARDS, HONORS & DISTINCTIONS

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|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| 05/2026           | <b>Best Teaching Assistant Award</b><br>Awarded by the Department of Electrical and Computer Engineering in recognition of outstanding teach-ing assistance, student support, and instructional contributions.                                                                                                                                                                                                                                  | Northwestern University              |
| 11/2025           | <b>Travel Grant</b><br>Awarded a departmental travel grant to present the paper "Caption-Driven Explainability: Probing CNNs for Bias via CLIP" at IEEE International Conference on Image Processing (ICIP) 2025 Conference in Anch-or-age, Alaska.                                                                                                                                                                                             | Northwestern University              |
| 10/2025           | <b>Hackathon Project: Deep Neural Operators for Detailed Binary Evolution Simulation</b><br>Developed a neural-operator-based approach for accelerating large-scale astrophysical simulations dur-ing the NCSA and NVIDIA hackathon, in collaboration with the POSYDON project and within the NSF-Simons AI Institute for the Sky (SkAI). The project was subsequently selected for presentation at the Open Accelerated Computing Summit 2025. | NVIDIA, NCSA                         |
| 09/2025           | <b>Medicus Scholarship</b><br>Awarded in recognition of academic excellence and cross-disciplinary research contributions as a Swiss doctoral student pursuing graduate studies in the United States. (USD 10,000)                                                                                                                                                                                                                              | Swiss Benevolent Society of New York |
| 09/2024 – 08/2025 | <b>Walter P. Murphy Fellowship</b><br>Endowed fellowship supporting outstanding PhD students during their first year of graduate study. (USD 45,000)                                                                                                                                                                                                                                                                                            | Northwestern University              |
| 07/2018           | <b>Red Dot Award–Winning Ophthalmic Surgery Device</b><br>Contributed to the engineering team at Stettbacher Signal Processing AG. Improved eye-tracking algo-rithms for robust off-center gaze detection and restored reliable Bluetooth file transfer functionality.                                                                                                                                                                          | Red Dot GmbH & Co. KG                |
| 01/2016           | <b>Recognition Award for Bachelor Project</b><br>Awarded for outstanding performance on a bachelor project conducted at Huber+Suhner AG. Developed a force-control system for a stranding machine to prevent corkscrew effects in custom cable production. (CHF 1,500)                                                                                                                                                                          | Huber+Suhner AG                      |
| 05/2003           | <b>Guinness World Records Participant</b><br>Participated in the Guinness World Records title for the largest simultaneous orienteering event (207,979 participants across 1,381 Swiss schools).                                                                                                                                                                                                                                                | Guinness World Records               |

## EXPERIENCE

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                    |
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| 07/2023 - 08/2024 | <b>Data Scientist / Management Team Member</b><br><ul style="list-style-type: none"> <li>Member of the management team; worked directly with CEO, CTO, and C-level executives on technical strategy and smart factory (Industry 4.0) initiatives in industrial sensing.</li> <li>Developed and deployed a machine learning model for estimating vehicle velocity in passive (unde-tectable) speed enforcement systems using difference brightness sensors under real-world constraints.</li> <li>Led the transfer of an explainable AI model for injection molding from research to production, resulting in integration into the ComoNeo system.</li> <li>Designed and deployed machine learning models for manufacturing process monitoring, reducing scrap rates and preventing costly production failures in support of Kistler AG's smart factory vision.</li> </ul> | Kistler AG         |
| 03/2023 - 06/2023 | <b>Traveller</b><br>Explored destinations across America, Canada, Colombia, Portugal, and Switzerland.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | All over the world |

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| 06/2017 - 02/2023 | <b>Engineer / Project Leader</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Stettbacher Signal Processing AG</b> |
|                   | <ul style="list-style-type: none"> <li>Developed a computer vision system (transfer learning with YOLOv5) with augmented visual overlays to detect and localize waste containers, enabling automated control of a robotic arm for bin emptying on waste disposal trucks developed by Villiger AG.</li> <li>Designed and implemented signal processing systems, including one of the most precise acoustic impedance measurement devices in the world used by EMPA.</li> <li>Contributed to the software stack (Linux, Bluetooth, eye tracking) of a Red Dot Award-winning ophthalmic device, improving eye-tracking robustness and system reliability.</li> <li>Led the development of a ruggedized tablet, coordinating international teams across time zones. Contributed to electronics, custom Linux system, and software. The device featured the brightest display available in a tablet at the time.</li> </ul> |                                         |
| 01/2020 - 02/2020 | <b>Soldier (World Economic Forum Deployment)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Swiss Armed Forces</b>               |
|                   | <ul style="list-style-type: none"> <li>Selected among Swiss Armed Forces personnel for deployment at the World Economic Forum Annual Meeting 2020 in Davos.</li> <li>Supported security and operational coordination for a high-profile international summit involving heads of state and global executives.</li> <li>Operated in a fast-paced, multinational environment requiring discipline, reliability, and situational awareness.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                         |
| 10/2016 - 05/2017 | <b>Development Engineer / Project Leader</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Optotune AG</b>                      |
|                   | <ul style="list-style-type: none"> <li>Selected to a 3-person team working directly with the CEO to drive a mission-critical initiative to scale liquid tunable lens technology for high-volume smartphone production; led a multidisciplinary team of 10 engineers (incl. PhD-level specialists) to develop a robotic production system, contributing to deployment in commercial devices (Xiaomi Fold 1 and 2).</li> <li>Designed and optimized control systems, increasing production yield by 20%, including work in clean-room environments.</li> </ul>                                                                                                                                                                                                                                                                                                                                                           |                                         |
| 08/2016 - 09/2016 | <b>Traveller</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Eastern Europe</b>                   |
|                   | Explored multiple countries across Eastern Europe, including Slovenia, Croatia, Serbia, Hungary, Czech Republic, and Austria.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                         |
| 08/2011 - 04/2016 | <b>Automation Engineer / Project Leader</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Comsys Bärtsch AG</b>                |
|                   | Developed software with PID controllers for home automation and smart home systems and trained apprentices.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                         |

## SKILLS

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| Core Expertise     | Explainable AI (XAI), Physics-Informed Neural Networks (PINNs), Neural Operators (PINOs), Scientific Machine Learning, Robustness & Domain Generalization, Signal Processing, Algorithm Development, Optimization |
| Robotics & Systems | Robotics, Embedded Systems, Real-Time Systems, Edge AI, Sensor Data Processing, Hardware-Software Co-Design, Control Systems                                                                                      |
| Programming        | Python (PyTorch, TensorFlow, Scikit-learn), C/C++, MATLAB, R                                                                                                                                                      |
| Tools              | Git, Linux, Conda, Jupyter, Docker                                                                                                                                                                                |
| Languages          | Swiss German (Native), German (Fluent), English (Fluent), French (Basic)                                                                                                                                          |

## PRESENTATIONS

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|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12/2025 | PhD qualification exam presentation on "Review of Physics-Informed Machine Learning (PIML)" at Northwestern University, evaluated by Prof. Aggelos K. Katsaggelos, Prof. Ermin Wei, and Prof. Ying Wu.                               |
| 09/2025 | Presented the paper "Caption-Driven Explainability: Probing CNNs for Bias via CLIP" at the IEEE International Conference on Image Processing Workshops (ICIPW) in Anchorage, Alaska (session honoring Prof. Aggelos K. Katsaggelos). |
| 06/2025 | Guest lecture on industry experience in the <i>Computational Optics</i> course at Northwestern University, instructed by Prof. Emma Alexander.                                                                                       |
| 03/2025 | Literature survey on "Robustness in Computer Vision" in the <i>Advanced Computer Vision</i> course at Northwestern University, taught by Prof. Ying Wu.                                                                              |
| 12/2024 | Delivered a lecture on "Unlocking Explainable AI: A Beginner's Guide" in the <i>Machine Learning</i> course at Northwestern University, taught by Prof. Aggelos K. Katsaggelos and instructed by Santiago Tapia.                     |

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| 12/2024 | Presented "Caption-based Explainable AI" in the <i>PhD Life Skills Building course</i> at Northwestern University, instructed by Prof. Nivedita Aurora.                                                                 |
| 11/2024 | Presented and discussed "Learning Transferable Visual Models from Natural Language Supervision" (Radford et al.) in the <i>Generative Deep Models course</i> at Northwestern University, taught by Prof. Bryan Pardo.   |
| 11/2023 | Presented the "AI in injection molding" project to the Plastics business unit at Kistler AG; the work was later integrated into the ComoNeo system.                                                                     |
| 07/2023 | Delivered a talk on interpretable machine learning at Kistler's Innovation Lab, introducing methods such as permutation importance, LIME, partial dependence plots, and Shapley values (materials available on GitHub). |
| 02/2023 | Presented the novel explainable AI method "Caption-based Explainable AI" at Northwestern University (in person) and the Eastern Switzerland University of Applied Sciences (online).                                    |
| 08/2022 | Presented a quantitative trading engine leveraging Gaussian processes for hyperparameter optimization at the Eastern Switzerland University of Applied Sciences.                                                        |
| 02/2022 | Presented an "AI in injection molding" anomaly detection system for product quality classification using cavity pressure curves at the Eastern Switzerland University of Applied Sciences.                              |
| 07/2016 | Presented the development of a medical device for brucellosis detection using lock-in signal processing at the Zurich University of Applied Sciences.                                                                   |
| 01/2016 | Presented a force-control system for stranding machines to prevent corkscrew effects in cable production at the Zurich University of Applied Sciences.                                                                  |

## SERVICE, TEACHING, AND OUTREACH

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| 05/2026 - Today   | <b>Visiting Scientist</b><br>Volunteer for Adler Planetarium's Astronomy Conversations program in the Space Visualization Lab. Discuss artificial intelligence, astrophysics, stellar evolution, and scientific machine learning with museum guests, helping make current research more accessible and engaging for visitors of all ages.                                                                                                                                                                                                   | <b>Adler Planetarium</b>                        |
| 05/2026           | <b>Invited Judge</b><br>Invited to serve as a judge for the Northwestern IEEE Technical Project Showcase. Evaluated undergraduate engineering projects across design, technical execution, impact, and innovation, providing feedback to 60+ students on projects spanning AI/ML, hardware, FPGA systems, embedded systems, accessibility, healthcare, education, and policy applications.                                                                                                                                                  | <b>Northwestern IEEE Student Branch</b>         |
| 04/2026           | <b>Peer Reviewer</b><br>Reviewed interdisciplinary research at the intersection of machine learning and digital art history (digital humanities), focusing on experimental design, model interpretability, and methodological rigor.                                                                                                                                                                                                                                                                                                        | <b>NPJ Heritage Science (Springer Nature)</b>   |
| 04/2026 - 09/2026 | <b>Local Organizing Committee Member</b><br>Invited to rejoin the local organizing committee in recognition of contributions to the inaugural conference. Supporting the planning, coordination, and execution of the 2026 edition, with continued focus on fostering collaboration between machine learning and astrophysics communities.                                                                                                                                                                                                  | <b>Open SkAI Conference 2026</b>                |
| 01/2026 - 03/2026 | <b>Teaching Assistant</b><br>Supported instruction for a graduate-level course in Scientific Machine Learning. Led office hours, assisted with assignments and projects, and mentored 60+ students on topics such as differential equations, optimization, physics-informed neural networks, and neural operators. Received the 2026 Best Teaching Assistant Award from the Department of Electrical and Computer Engineering.                                                                                                              | <b>Northwestern University</b>                  |
| 04/2025 - 09/2025 | <b>Local Organizing Committee Member</b><br>Served on the local organizing committee and created the "AI + Astrophysics Glossary," a collaborative reference bridging technical vocabulary across machine learning and astrophysics. Supported the planning, logistics, and communication for the first Open SkAI Conference.                                                                                                                                                                                                               | <b>Open SkAI Conference 2025</b>                |
| 06/2025 - 08/2025 | <b>Research mentor</b><br>Selected as a mentor for the highly competitive Student Research Institute (over 2,000 applicants; 70 admitted; 3.88% acceptance rate), supporting students from underserved backgrounds with little prior research experience. Mentored Shaurya Jeevagan on a full physics-informed neural network (PINN) project involving advection-diffusion modeling, Fourier feature construction, boundary-condition handling, PDE-residual sampling, and neural-network training with data-physics hybrid loss functions. | <b>Harvard Undergraduate OpenBio Laboratory</b> |
| 09/2023 - 06/2024 | <b>Master thesis mentor</b><br>Mentoring Ken Geeler over the course of his master thesis classifying the correct air pressure in tires using a machine learning model and Kistler AG's Weight In Motion sensor.                                                                                                                                                                                                                                                                                                                             | <b>Kistler AG</b>                               |

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| 11/2019 - 04/2021 | <b>Volunteer coach</b><br>Coaching refugee Sharif Shayan during his internship at UBS (largest Swiss bank), which led to a permanent position at Brack.ch (Swiss e-commerce company). | Powercoders |
| 03/2013 - 01/2016 | <b>Math tutor</b><br>Math tutoring for Robin Kreis, Nicolas Kleinert, Roman Kümpel and Sandro Rovetto.                                                                                | Freelancer  |